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THE STATIONARY CRETACEOUS PALEOMAGNETIC POLE OF SICHUAN (SOUTH CHINA BLOCK)

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Abstract. Upper Jurassic to lower Tertiary sites have been sampled from three localities, separated by up to 150 km, in the Sichuan Basin (~30°N, 103°E) of the South China Block. Each locality has at least one stability test, but none alone gives a fully reliable result. When taken together, however, identical directions are found with a global average D=12.7°, I=34.2°, k=42.9°, α_{95}=3.3° (N=44 sites) and corresponding pole 73.9°N, 234.4°E, dp=2.2°, dm=3.8°, paleolatitude λ=18.8°. There is no significant apparent polar wander (APW) with age, which is consistent with an acquisition of magnetization during the period of the stable tip of the Eurasian APW loop (~130-70 Ma), as defined with poles from other continents transferred onto Eurasia. This age range is compatible with the available paleontological and magnetostratigraphic age constraints but is shorter than that suggested by the assigned formation ages. The paleolatitude of Sichuan is significantly lower (8.0°±4.0° at 90 Ma) than that predicted by the reference APW path, either because of northward shortening during the Tertiary or because of errors in the APW path. Cretaceous poles from other Chinese blocks are consistent with the Sichuan pole, but there is evidence that these blocks have suffered local rotations up to 10° during the Tertiary, probably as a result of the India-Eurasia collision.

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INTRODUCTION

Throughout the Phanerozoic, continental blocks have accreted onto the southeast frontier of Asia [e.g., Zhang et al., 1984]. The Indian subcontinent was the latest block to collide and, in the process, substantially deformed and reorganized the region [e.g., Tapponnier et al., 1986] to form its present configuration (Figure 1). The timing of accretion and exact geometry of these blocks is hard to establish, but paleomagnetism has been one of the tools used to help this endeavor. Poles postdating the collisions can be used to study the internal deformation of the continent, and in the case of no deformation these poles can be used to define the reference apparent polar wander path (APWP) for the whole continent.

In this paper we examine the Cretaceous paleomagnetic record of the South China Block, as recorded by the redbeds of the Sichuan Basin. This age is of interest because it probably postdates the accretion events of China but predates the Indian collision. Most of the major Chinese blocks have reliable poles of similar ages, and thus detailed comparisons can be made. Unfortunately, reliable Cretaceous poles from other parts of Eurasia are scarce, so comparisons must be made with poles from other continents transferred onto Eurasia using ocean kinematics.

Discussion has taken place in the literature about the ~15° misfit between southeast Asian Cretaceous poles and those from "stable" Eurasia [e.g., Lin et al., 1985; Courtillot and Besse, 1986]. Lee et al. [1987] suggested that the Eurasian reference APWP was in error, and Besse and Courtillot [1991] developed a new APWP using reliable poles from Africa, North America, Eurasia (including China), and India. The Chinese poles fit rather well with the new reference APWP for Eurasia. However, it is of interest to cast an even finer look on paleomagnetic results from China, and in this paper we will reexamine the relationship between this

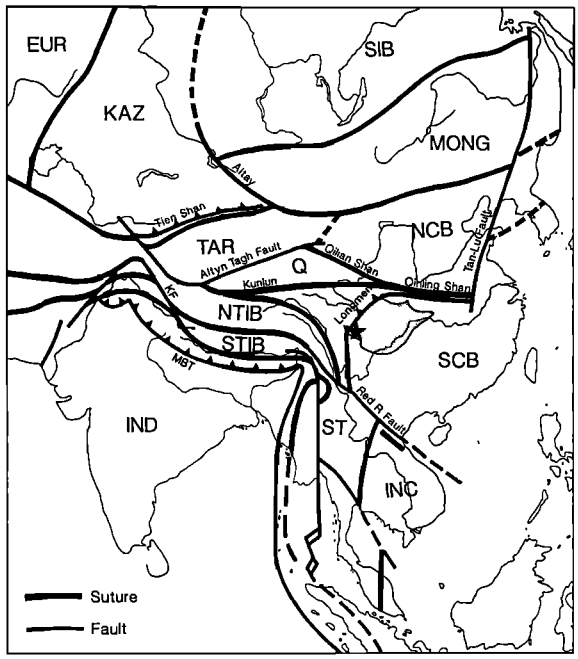


Fig. 1. Schematic tectonic map of China and southeast Asia, showing sutures and major faults. The major blocks are EUR, Europe; KAZ, Kazakhstan; SIB, Siberia; TAR, Tarim; NCB, North China Block; SCB, South China Block; Q, Qaidam; NTIB, North Tibet; STIB, South Tibet; ST, Shan Thai; INC, Indochina; and IND, India. The Sichuan Basin is the stippled region, and the sampling area is marked with a star. Based on Li et al. [1982] and P.Tapponnier (personal communication, 1989).

region and the rest of the world. Despite uncertainties in the precise ages of Cretaceous formations in China the geometry of the region, its poles, and the APWP allow a useful analysis.

GEOLOGY, FIELD WORK AND LABORATORY WORK

Under Sino-French cooperation we held two fieldtrips to the Sichuan Basin during November 1986 and November 1988. During both trips we sampled sites going back weli into the Mesozoic, but in this paper we will only consider results from formations dated from the Upper Jurassic to the lower Tertiary.

The Sichuan Basin is a large sedimentary basin mostly filled with Mesozoic continental redbeds, but underlying marine sediments dating up to the early Upper Triassic outcrop on the basin's border and, most notably, along long anticlinal hills along the southeast part of the basin (Figure 2). Cretaceous rock crops out along the northwest, west and south of the basin. Most of it is gently dipping, but folding relating to the Longmen Shan thrusts on the western border of the basin is evident. We have sampled both folded and relatively flat-lying localities, giving us the possibility of a significant fold test and a check against local rotations brought about by tectonic disturbances.

Geological dating and stratigraphy in these continental red sandstones is difficult. Paleontological ages are based on (scarce) continental ostracods which have few distinguishing marks. Usually only the genus and not the species of these fossils was identified. We found no ostracods from our collection, and we treated eight of our samples to look for pollens but only found a few unidentifiable wood remains. As an example of the problems encountered [Sichuan Bureau of Geology, 1982, p.165],

> Near the Longmen Shan, the Tianma Shan formation has a hidden disconformity with the Penglai Zhen formation below. It is easy to distinguish the limit as there is a conglomerate layer at the bottom of the Tianma Shan formation. On the other hand, near Ya'an, Ming Shan and Dayi, the two formations have a transitional conformity, and in the case of missing fossils the limit is difficult to define. At present, the Tianma Shan formation is separated from its underlying formation (Penglai Zhen) by its clear colour and a thick and stable conglomerate layer. In general, the two formations have a confirmed contact only in some localities, a hidden disconformity in others.

Li [1988] makes it clear that even in type sections in this region, correlation with the European time scale is debatable. Accordingly, we are forced to use paleomagnetic as well as geological age indicators in this study.

In 1986 we collected Cretaceous rocks from three localities, and in 1988 we collected from one more Cretaceous locality (Figure 2, Table 1). One section, taken along a river south of the city of Ya'an from the village of Saping (29.9°N, 103.0°E), with assigned ages from the Lower Jurassic (Ziliujing formation) to Lower Cretaceous (Jiaguan formation) has a complicated structure and was mostly remagnetized, and thus we will not consider this section further in this paper. Details of this section are given in Enkin [1990].

The sites (ZS21-ZS43) in the river section around the village of Feixianguan (30.0°N, 102.9°E), 10 km west of the city Ya'an, are described in detail elsewhere [Enkin et al., 1991]. The lithology was relatively uniform fine red sandstone throughout the 3 km of stratigraphic section sampled, with assigned ages from the Upper Jurassic (Penglai Zhen formation) to the lower Tertiary (Lu Shan formation).

The section lies just east of the Longmen Shan thrust system (marked A in Figure 3). Most sites sit on one limb of a syncline, but two sites were taken close to the thrust fault bounding the east side of the syncline. The fault motion set these sites almost vertical, providing the possibility of a fold test. Further north this fault reduces to an anticlinal fold which terminates ~30 km from the sampled section. This geometry indicates that the block has suffered a local counterclockwise (ccw) rotation. We [Enkin et al., 1991] estimated the magnitude of rotation by noting that the trend of most faults and fold axes within ~30 km of the sampling site is about 15° different from the trend of the site's fold axis and bordering fault. The underlying assump-

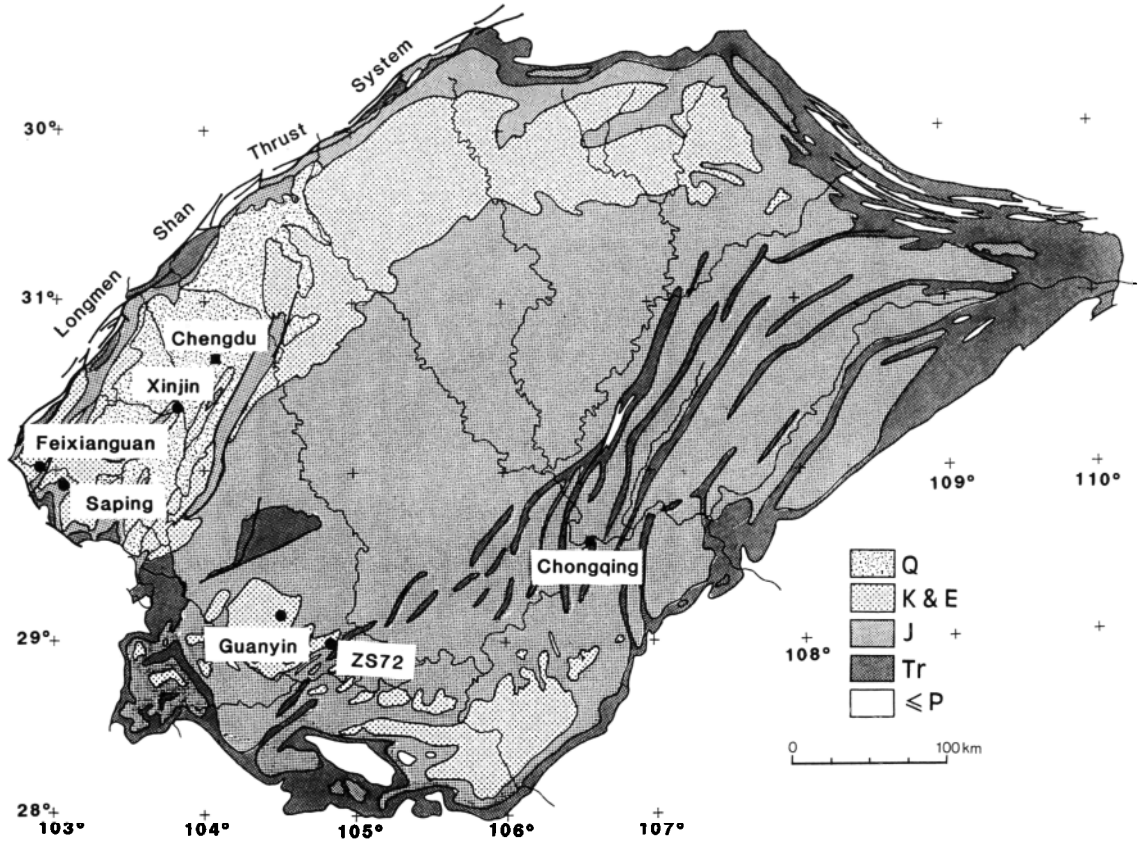


Fig. 2. Geological map of the Sichuan Basin. The sampling localities are marked with circles. Based on the geological map of Sichuan, scale 1:1,000,000 [Sichuan Bureau of Geology, 1983].

TABLE 1. Geological Formations and Reported Site Ages

		Locality / Site Numbers		
Code	Formation Name	Feixianguan	Xinjin	Guanyin
E2l	Lu Shan	42,43		
E2y	Yuguang Po	41		
K2-E1j	Jingi Guan	37-40		
K2gk	Guankou (top)	36		59-64,70,71
	(bottom)	31-35b*		
K1j K2s	Jiaguan Sanke	26-30	44,45,47	69
K1d	Daerdang			65-68
K1t	Tianma Shan	25	46	
J3p	Penglai Zhen	21-24		72

Codes are J3, Upper Jurassic; K1, K2, Lower and Upper Cretaceous; E1, Paleocene; E2, Eocene. Subscripts to the age code identify the formation.

*Sites 30 to 35b are magnetostratigraphically dated at just before the Cretaceous Long Normal Superchron and thus represent a Lower Cretaceous (K1) age.

This section was almost completely flatlying, but a fourteenth site (ZS72), of Upper Jurassic age, was collected on another point (~50 km SE) bordering this Cretaceous body, and we include it in this study. These sites often consisted of a single firm layer bounded by red mud. These rocks alter so quickly that only fresh quarries or construction cuts could be sampled.

At each site, eight to sixteen standard 2.5 cm diameter cores were taken with electric or gasoline powered drills. Orientations were taken with magnetic and when possible with sun compass (only three sites because of rather unfriendly weather). Bedding directions were taken at three to twelve places at each site, being careful to avoid cross bedding which was observed at all scales up to tens of meters.

Most samples were demagnetized thermally, as alternating field demagnetization never efficiently isolated magnetic components. At least 10 demagnetization steps were done on each sample. Samples were measured in Paris with a CTF three-axis cryogenic magnetometer and in Beijing with a Schönstedt fluxgate spinner. Directions and great-circle demagnetization paths were isolated using principal component analysis [Kirschvink, 1980], and site averages were found with McFadden and McElhinny's [1988] method.

RESULTS

Each of the three localities studied has several peculiarities which deserve to be discussed separately. The fact that the magnetic properties are similar within each section but dissimilar between sections is an indication that the final directions, since we find them to be the same, are independent of lithology and environmental history.

The results from the Feixianguan section are published in a separate paper [Enkin et al., 1991], so here we shall just give a summary. The 28 sites (ZS21 to ZS43) of this section all revealed two components: a well-defined high-temperature component (HTC) (typically 500°-680°C) and a present Earth field (PEF) direction Low-Temperature Component (LTC) (NRM to ~500°C). Both polarities are represented among the HTCs with about two thirds of the samples in normal polarity. Although the demagnetization curves were usually rather noisy, directions or great circles could almost always be defined, and 86% of the 450 specimens measured were used in the final averages. The fold test is significantly positive (99%) even though it was realized with only two sites which have significantly different bedding from the rest.

Sites ZS30 to ZS35B (assigned to the Upper Cretaceous) were continuously and densely sampled, and the magnetostratigraphic profile, featuring five reversals, correlates well with M4 to M0 (i.e., the period just before the Cretaceous Long Normal Superchron). While these sites must therefore be reassigned to the Lower Cretaceous, extrapolation to the other sites suggests that the assigned ages of the other sites are not unreasonable.

The assigned ages (Table 1) go from the Upper Jurassic to the Eocene. To our surprise, the paleomagnetic direction recorded by this 3 km thick sedimentary column is remarkably stationary, so a single average direction was taken for the whole series. Two sites were eliminated from the locality average, one (ZS25) because it was poorly defined, the other (ZS42) because it was a clear outlier (still not understood).

The samples of the Xinjin locality had generally very well defined demagnetization paths, nearly always with two clear components (e.g., Figure 5, 45-12B and 46-02B). The remarkable aspect of the magnetic behavior of these rocks was the dispersion of the LTC directions, which seem to be distributed uniformly over the sphere.

The HTC, on the other hand, was quite consistent within each site (Table 2), and the clustering of the locality mean is tighter after structural correction, but the fold test is not confirmed with only four sites (Figure 6, Table 2). Site ZS46 is magnetized in the reverse direction, very close to the antipode of the other three sites, and thus we consider the remanence of the sites from Xinjin to be primary.

The Cretaceous sites of the Guanyin locality (Table 3) were all magnetized with normal polarity (e.g., Figure 5, 59-01B and 63-09B), except for a few poorly behaved reversed polarity samples from ZS68 and ZS69. Assuming these anomalous samples are merely the result of in-laboratory remagnetization, the consistent normal polarity probably means that all these sites were deposited during the Cretaceous Long Normal Superchron. The LTC and HTC were often difficult to distinguish. The average direction of 89 LTCs is D=19.9°, I=47.7°, k=3.8, α_{95}=9.0° in geographic coordinates, not in the PEF direction (D=0°, I=49°) but on the great circle between the PEF and the final direction.

The HTC was usually well defined except for sites ZS65, ZS66, ZS68, and ZS69 which were more dispersed (α_{95}>15°). The susceptibility of these rocks was measured, and only these four sites had Koenigsberger ratios under 1.0, implying that their magnetic carriers are less efficiently magnetized than the rest. Nevertheless, there did not seem to be any strong reason to eliminate any of these sites, since they were not outliers in the site distribution (Figure 7).

The fold test cannot be significant in a region of such flatlying strata, but the scattering is slightly worse in stratigraphic than geographic coordinates. It would be difficult to accept this direction as primary if we were not able to compare this direction with that of site ZS72 (e.g., Figure 5, 72-09B), which is antipodal and comes from a region well separated from the rest. The immobile paleodirection throughout the Feixianguan section justifies the comparison of different age directions in the Guanyin section.

DISCUSSION

Fisher averages of several groupings of these sites are given in Table 4. Because of the range of structures found in this study, only the McElhinny [1964] fold test is reported. An application of the McFadden and Jones [1981] fold test to the Feixianguan section is given by Enkin et al. [1991]. First we will discuss the evolution of the paleomagnetic direction with the age of the rocks, and then we will study the localities both separately and combined.

To study the variation of the paleodirection with age we usually accepted the assigned ages, with the exception of sites ZS30 to ZS35B which are magnetostratigraphically

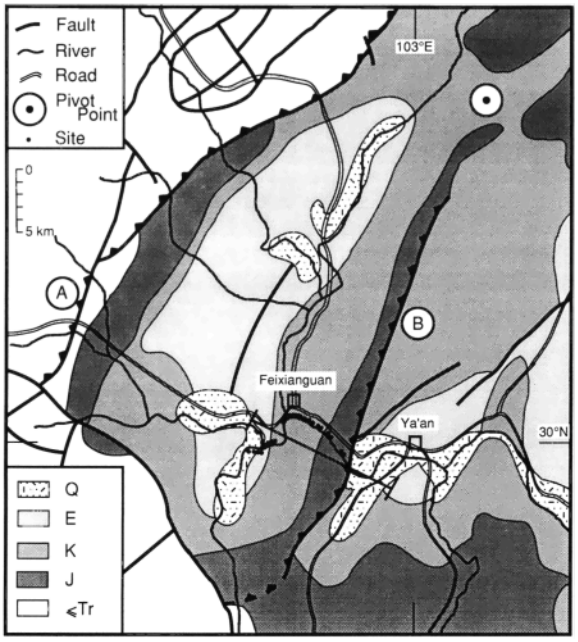


Fig. 3. Geological map in the region of Feixianguan. Rotation around the pivot point is estimated by determining the amount of shortening along the marked section south of the Ya'an-Feixianguan road (see Figure 4).

tion is that the structures started parallel under a uniform direction of maximum stress. As the Longmen Shan thrust system developed a corner in the vicinity of the Feixianguan syncline, edge effects changed the direction of maximum stress and rotated the syncline.

An independent method to determine the amount of rotation is to estimate the amount of shortening along the section (Figure 4). While the available data are insufficient to rigorously balance the section (e.g., the depth of decollement is unknown), we are able to construct a section which is compatible with our field observations, the geological map [Sichuan Bureau of Geology, 1983] and the Landsat photograph (given in Figure 18 of Enkin et al. [1991]). Details of the balanced section are given in Figure 4. We find 4 km of shortening, 30 km from the pivot point, giving ~8° of ccw rotation. This quantity of shortening is a lower bound since unobserved deformations or faults can increase the value. In the discussion that follows, we will present data with both 8° and 15° of local rotation, each with an estimated uncertainty of 5°, but we favor the former, more rigorous value.

To check for the effects of tectonic rotations and remagnetization which could be caused by the nearby Longmen Shan Thrust system, we also sampled, in 1986, four Lower Cretaceous sites around the city of Xinjin (30.4°N, 103.8°E) not far from Chengdu (Figure 2). Sites ZS44, ZS45, and ZS47 are from quarries, and ZS46 was a small natural outcrop. The sandstone is very coarse, red, and soft. Site ZS44 is reported to have the fish fossil *Lepidotes* [Xu Xingxi, personal communication, 1986], typical of the Lower Cretaceous Jia Guan formation [Sichuan Bureau of Geology, 1982, p.169].

Finally, in 1988, we sampled from thirteen Cretaceous sites (ZS59-ZS71) around the village of Guanyin (29.1°N, 104.6°E; Figure 2). The sites are assigned to the Lower Cretaceous Daerdang formation and the Upper Cretaceous Sanhe and Guankou formations. There is controversy of the correlation of these units with those defined further west in the basin [Sichuan Bureau of Geology, 1982, p.185], but without further information we will use the assigned ages.

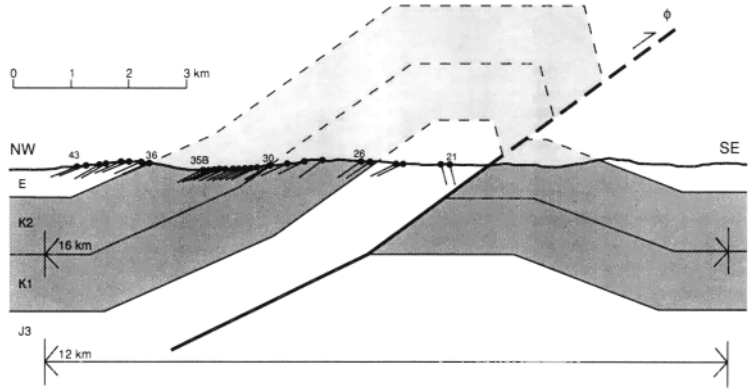


Fig. 4. Simplified balanced section of the Feixianguan sampling area. The constraints on the section leave little room for significant modification. The bedding dip at each sampled site is marked. The level separating formations assigned to K2 and E ages is a hard layer which can be identified clearly on the Landsat photograph (cf. Enkin et al. [1991], figure 18). The fold axis of the anticline does not outcrop, and there is the possibility of a fault there, which would allow considerably more shortening. The section which is now 12 km long was about 16 km long when it was flat, indicating 4 km of shortening.

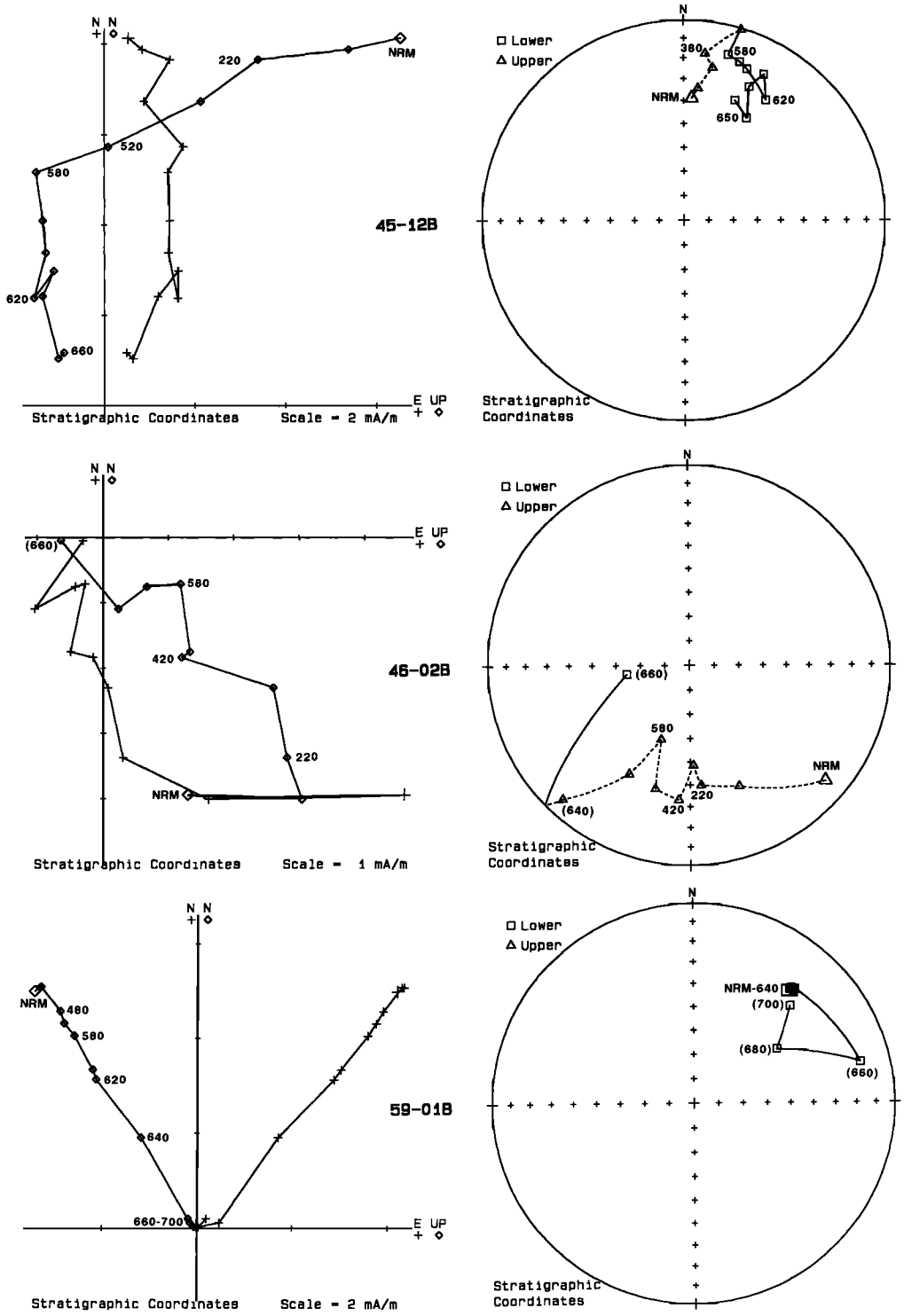


Fig. 5. Representative vector Zijderveld and stereographic diagrams (in stratigraphic coordinates) of specimen demagnetizations.

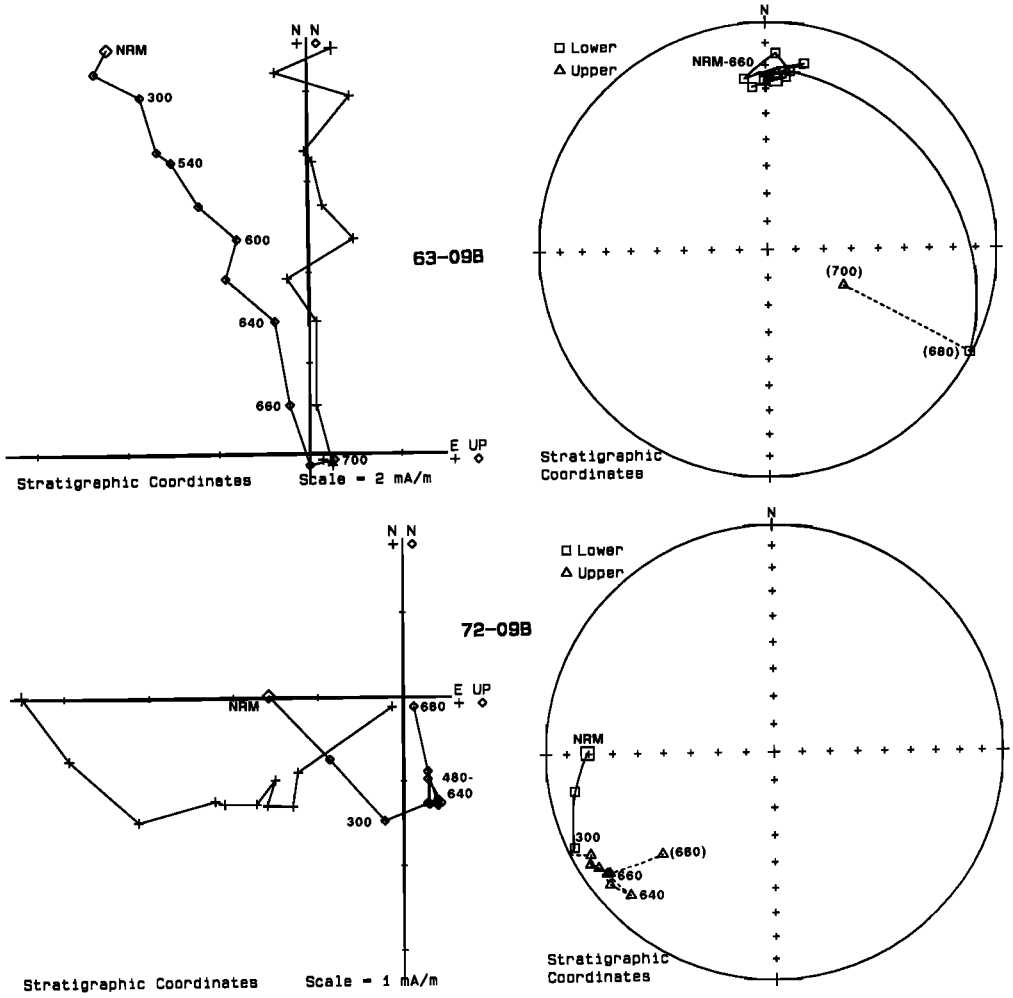


Fig. 5 (continued)

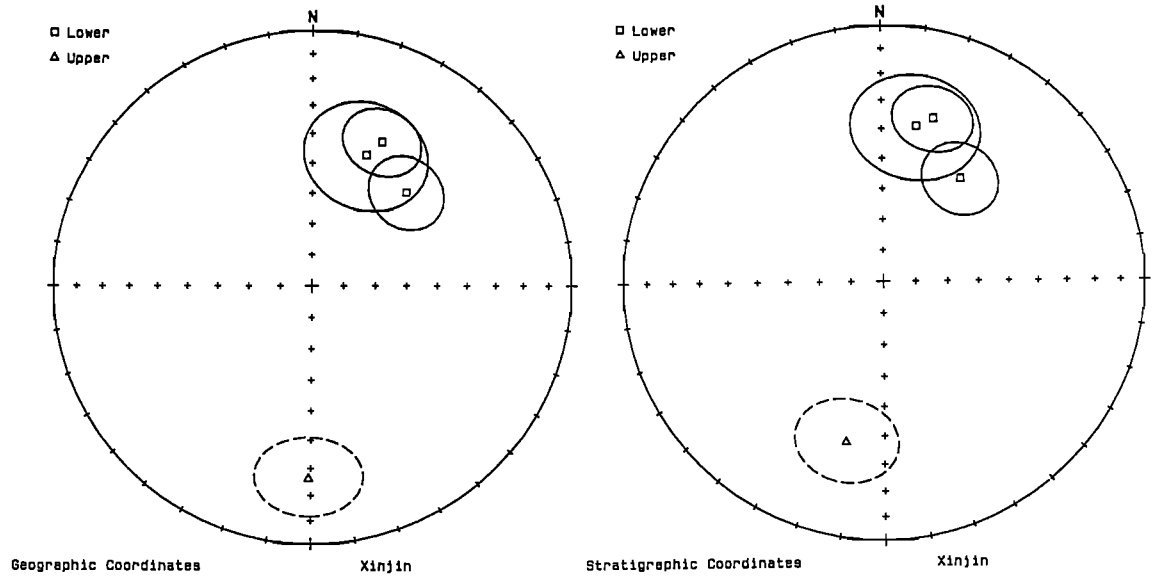


Fig. 6. Site mean directions of Xinjin locality sites.

TABLE 2. Site Data for Xinjin Section (30.4°N, 103.8°E)

Site	Form'n	S,D	Specim	NRM, mA/m	MDT, °C	Nor, percent	D_g, deg	I_g, deg	D_s, deg	I_s, deg	k	α_{95}, deg
44	K1j	213,12	10/11	4.1	620	100	25.6	37.5	17.3	33.9	19.1	11.5
45	K1j	224,13	10/12	5.5	620	100	22.2	43.8	12.3	37.9	7.9	18.4
46	K1t	32,22	8/12	1.1	400	0	180.7	-26.3	193.4	-36.4	20.1	14.7
47	K1j	210, 7	7/12	2.4	620	100	44.8	47.1	36.8	48.5	28.6	11.8

Columns are Site, site number; Form'n, formation code (as in Table 1); S,D, strike and dip in degrees; Specim, number of specimens used in average / number of specimens measured; NRM, natural remanent magnetization intensity; MDT, approximate median destructive temperature; Nor, proportion of normal polarity samples, D, declination; I, inclination; subscripts g and s,, geographic and stratigraphic coordinates respectively; k and α_{95}, Fisher distribution parameters.

TABLE 3. Site Data for Guanyin Section (29.1°N, 104.6°E)

Site	Form'n	S,D	Specim	NRM, mA/m	MDT, °C	Nor, percent	D_g, deg	I_g, deg	D_s, deg	I_s, deg	k	α_{95}, deg
59	K2gk	285,3	12/12	5.2	640	100	43.5	35.4	42.4	32.5	57.8	7.3
60	K2gk	332,6	10/11	1.8	640	100	28.2	33.3	30.0	28.6	57.5	8.9
61	K2gk	252,5	11/11	5.6	620	100	352.5	33.4	351.9	28.5	20.5	12.5
62	K2gk	252,5	12/12	8.6	500	100	12.0	16.9	11.3	12.6	50.6	7.9
63	K2gk	240,8	11/11	8.7	620	100	4.3	37.2	1.4	30.3	42.2	9.4
64	K2gk	240,8	11/11	18.4	600	100	17.9	35.1	14.2	29.4	27.8	11.7
65	K1d	0,0	7/11	1.3	560	100	12.1	32.4	12.1	32.4	15.6	17.5
66	K1d	0,0	9/12	1.1	500	100	358.4	39.7	358.4	39.7	12.3	28.8
67	K1d	207,12	12/13	5.5	440	100	27.8	54.1	12.3	52.6	97.4	5.2
68	K1d	209,13	11/13	2.7	440	85	9.6	49.6	357.0	43.9	11.1	19.7
69	K2s	194,5	11/12	1.7	440	83	15.7	27.9	13.1	27.9	16.5	15.6
70	K2gk	192,7	10/10	23.3	620	100	29.4	30.1	25.1	32.0	21.9	13.2
71	K2gk	244,2	13/13	21.9	460	100	35.9	24.2	33.2	20.4	28.4	9.8
72	J3p	200,12	9/17	8.5	260	24	199.6	-21.3	195.0	-20.8	50.7	10.4

Same columns as Table 2.

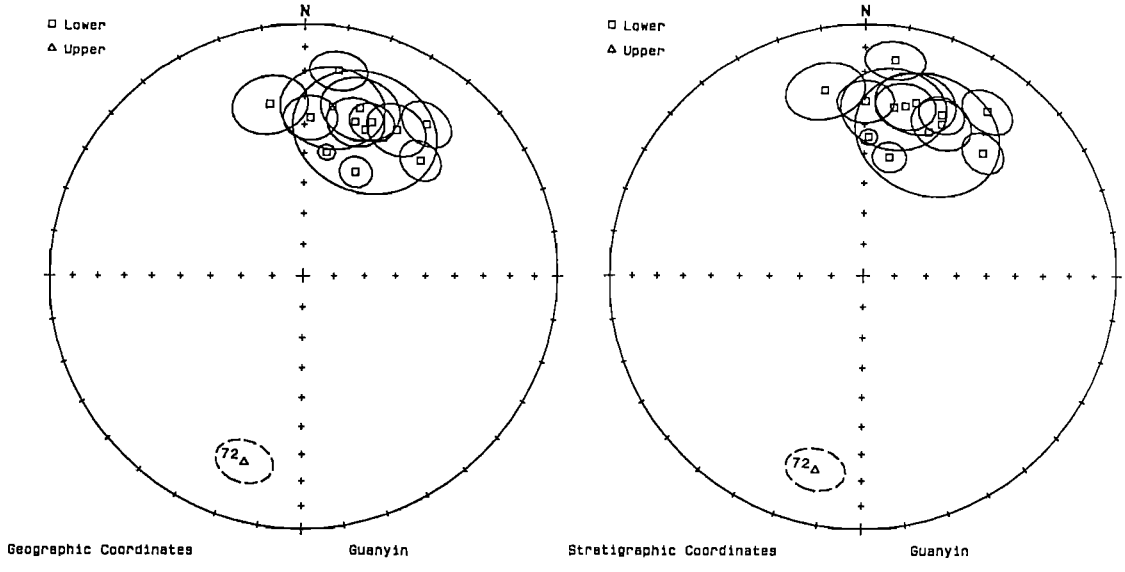
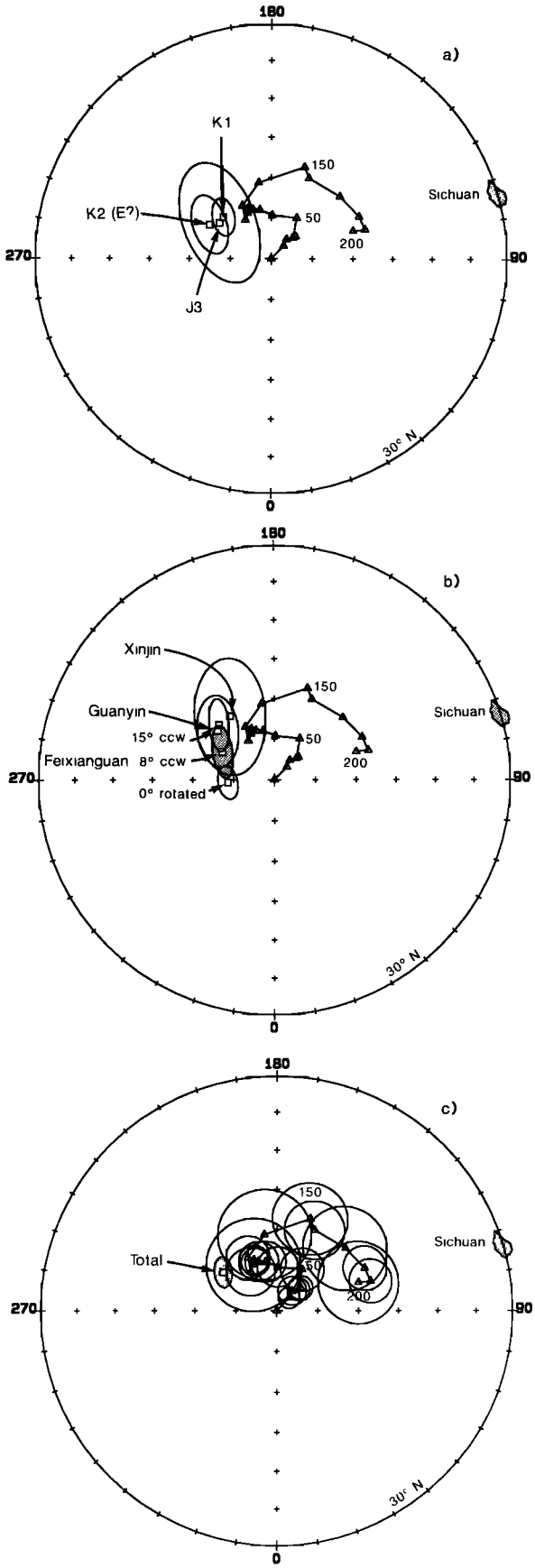


Fig. 7. Site mean directions of Guanyin locality sites.

TABLE 4. Fisher Averages of Site Groupings

Collection	N	n	Geographic Coordinates D_G, deg	I_G, deg	k	α_{95}, deg	Stratigraphic Coordinates D_S, deg	I_S, deg	k	α_{95}, deg	k_S/k_G	λ, deg	VGP lat, deg	long, deg	dp, deg	dm, deg
By Age:																
K2-E	0°	16	189	16.6	33.5	40.2	5.9	8.4	31.2	25.7	7.4	0.77	16.8	74.8	250.8	4.6	8.3
	8°							11.9	30.9	32.6	6.6	0.81	16.7	72.8	241.1	4.1	7.4
	15°							15.0	30.8	38.0	6.1	1.1	16.6	70.8	234	3.8	6.8
K1	0°	23	305	21.8	37.0	42.8	4.7	8.0	36.5	50.6	4.3	1.2	20.3	77.9	244.5	2.9	5.0
	8°							13.3	36.4	57.5	4.0	1.3	20.2	74.5	229.0	2.7	4.7
	15°							18.0	36.4	56.7	4.1	1.3	20.2	71.0	219.8	2.8	4.8
J3	0°	5	62	6.9	16.3	8.4	28.0	5.9	34.8	32.9	13.5	3.9!	19.2	77.9	255.4	8.9	15.5
	8°							12.0	34.7	34.2	13.2	4.1!	19.1	74.6	235.3	8.7	15.5
	15°							17.4	34.8	33.3	13.5	4.0!	19.2	70.9	223.5	8.9	15.5
By Polarity:																	
Normal	0°	32	402	20.7	33.3	20.6	5.7	12.2	35.1	38.5	4.2	1.9!!	19.4	74.7	234.1	2.8	4.8
	8°							16.2	35.0	42.2	4.0	2.0!!	19.3	71.8	225.4	2.7	4.6
	15°							19.7	35.0	41.0	4.0	2.0!!	19.3	69.2	219.4	2.7	4.6
Reverse	0°	12	154	191.3	-34.2	84.8	4.7	177.0	-31.9	71.4	5.2	0.84	17.3	77.0	295.8	3.3	5.8
	8°							183.6	-31.8	99.8	4.4	1.2	17.2	76.8	267.8	2.8	4.9
	15°							189.4	-31.7	120.9	4.0	1.4	17.2	74.6	247.1	2.5	4.5
By Locality:																	
Feixian-guan	0°	26	372	17.1	32.0	23.0	6.0	2.0	34.2	63.1	3.6	2.7!!	18.8	78.6	273.4	2.4	4.1
	8°							10.0	34.2	63.1	3.6	2.7!!	18.8	75.6	241.8	2.4	6.5
	15°							17.0	34.2	63.1	3.6	2.7!!	18.8	70.9	225.2	2.4	6.5
Xinjin	4	35	21.8	39.8	23.4	19.4	19.2	39.2	58.5	12.1	2.5	22.2	71.0	214.8	8.6	14.5	
Guanyin	14	149	18.9	34.9	28.6	7.6	15.8	32.7	26.7	7.8	0.93	17.8	71.7	229.1	5.0	8.8	
Guanyin, (No ZS72)	13	140	19.1	35.8	27.7	8.0	16.1	33.5	25.6	8.3	0.92	18.3	71.8	227.3	5.4	9.4	
Xinjin and Guanyin	18	184	19.5	36.0	28.5	6.6	16.5	34.2	30.3	6.4	1.1	18.8	71.3	226.1	4.2	7.3	
Total (Feixianguan, Xinjin and Guanyin):																	
	0°	44	556	18.1	33.6	25.1	4.4	7.9	34.4	36.4	3.6	1.5!	18.9	76.8	248.3	2.4	4.1
	8°							12.7	34.2	42.9	3.3	1.7!!	18.8	73.9	234.4	2.2	3.8
	15°							16.8	34.2	44.9	3.2	2.0!!	18.8	71.1	225.5	2.1	3.7

Site means were flipped to the same polarity for the averages. The 0°, 8°, or 15° in the collection column marks the amount of counterclockwise rotation applied to sites from the Feixianguan section. Columns are N, number of sites; n, number of specimens; D, declination; I, inclination; k, precision parameter; α_{95}, 95% significance angle; subscripts G and S, geographic and stratigraphic coordinates, respectively; k_S/k_G, ratio of precision parameters in the two coordinate systems with fold test !, 95% and !!, 99% significance, respectively; λ, paleolatitude; lat, latitude; long, longitude; and dp, dm, semiaxes of VGP uncertainty. The underlined direction is the preferred Cretaceous average for the Sichuan Basin.



dated to the Lower Cretaceous despite their Upper Cretaceous assignment. The sites were separated into three age groups: the Upper Jurassic, the Lower Cretaceous, and the Upper Cretaceous and lower Tertiary combined. Only two sites, ZS41 and ZS43 (since ZS42 is rejected as an outlier), are from formations assigned to uniquely Tertiary ages (see Table 1). Since their directions are clearly similar to the Upper Cretaceous directions, we included them in the last age group.

The Upper Jurassic group, which includes the steeply dipping sites ZS21 and ZS22, significantly passes the fold test (with 95% confidence). While the other age groups are not sufficiently folded for a significant fold test, we note that each age group contains both polarities and consists of sites from at least two sections. These facts favor the interpretation that the determined directions are primary.

Although the collection could document up to 100 m.y. of apparent polar wander, there is remarkably little motion up the column (Figure 8a). This is particularly noteworthy in the Feixianguan section where 3 km of rocks spanning the entire age range show no evolution of the direction [cf. Enkin et al., 1991, Figure 11]. Accordingly, directions of all ages have been averaged together to study regional differences. Note also that since the paleomagnetic direction is constant through time, one need not worry too much about the correlation of formations between localities.

The normal and reversed polarity sites taken separately are close to but significantly different from antipodal (169.0°±5.9°, using 8° rotation of Feixianguan section). As discussed by Enkin et al. [1991], very few reversed directions were directly determined so that intersection of great circles was the only way to arrive at the reversed site means. Since most reversed sites lie on one monoclinal limb (Feixianguan section) and the PEF component is always rather well defined, the demagnetization paths tend to be subparallel, and therefore bias [McFadden and McElhinny, 1988] pushed the reversed means away from their true positions.

The individual localities have been discussed in the results section above. Figure 8b shows that their average poles are very similar (see also Table 4). We note that a 15° rotation of the Feixianguan section brings the three localities into better correspondence than the 8° rotation (which we favor, but note again to be a lower bound). However, even with 8° rotation the ellipses of uncertainty all overlap, indicating that the differences are not significant.

The final average of the 44 good sites from three separated localities is highly clustered and has a significant positive fold test (99%). We consider this total direction (in-

Fig. 8. Virtual geomagnetic poles determined in this study for: (a) different geological ages (all localities averaged for each age window); (b) different localities (all ages averaged for each locality; in the case of the Feixianguan section, poles are given under the hypothesis of no local rotation, 8° counterclockwise and 15° counterclockwise rotation (see text)); and (c) all sites taken together. In all cases (except for the unrotated Feixianguan direction), there are no significant differences between poles. The Eurasian apparent polar wander path is marked in triangles with ages in Ma (with 95% uncertainties drawn in Figure 8c). The Sichuan Basin is drawn around 30°N, 105°E.

cluding an 8° rotation of the Feixianguan section) to be our most reliable for the Cretaceous of the Sichuan Basin (Table 4, Figure 8c). This pole supersedes the pole isolated by Otofuji et al. [1990] based on 27 hand samples mostly from the Feixianguan section. A critique of that study is offered by Enkin et al. [1991].

The Sichuan pole matches very well with other Cretaceous poles from around China (Figure 9). The only outliers are the Upper Cretaceous poles of Nanjing and South Sichuan (only 12 samples) [Kent et al., 1986]. We will discuss the Nanjing pole later. The pole from South Sichuan (which is bounded by very tectonically active zones) has the same paleolatitude as the others but has been rotated by about 15° ccw. K. Huang and N. D. Opdyke (personal communication, 1990) have resampled this locality and find the same paleolatitude but a much smaller rotation. We conclude that the blocks of China were assembled roughly in their present configuration by at least the Late Jurassic. Note that the extrusion model of Tapponnier et al. [1986] predicts that South China should have moved with respect to North China along the Qinling fault belt since the India-Eurasia collision. This motion is not resolvable by paleomagnetism, since the mostly east-west orientation of this fault leads to a longitudinal shift, and thus the predicted rotation is quite small.

The Tarim Basin [Li et al., 1988] and the Junggar Basin [Chen et al., 1991] show no significant evolution of the paleomagnetic pole from the Jurassic-Cretaceous boundary to the Cretaceous-Tertiary boundary. The present study of the Sichuan Basin is the first place in China where dense sampling through the Cretaceous has been carried out, and here we see no significant APW at any time through this period. Note that Li et al. [1988] obtained a fold test on two monoclinal series of different ages and thus implicitly assumed no motion of the pole, a hypothesis now vindicated by the present study.

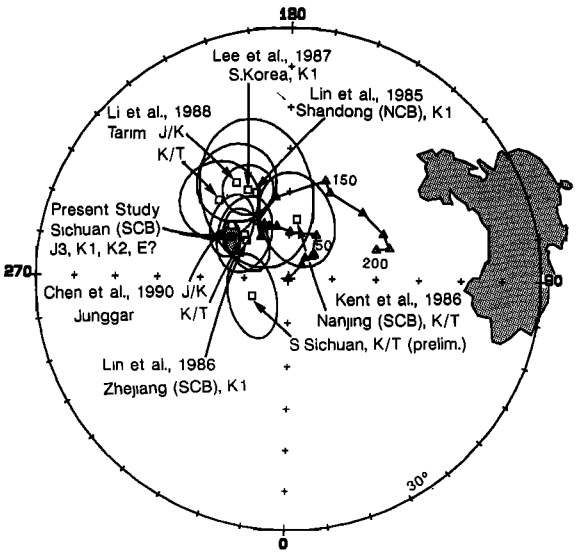


Fig. 9. Published Cretaceous poles from China. The Eurasian apparent polar wander path is marked in triangles with ages in Ma. The outline of China helps in locating the approximate position of the sampling sites.

In order to compare the Chinese Cretaceous poles with the Eurasian reference APWP we took the same pole list as used by Besse and Courtillot [1991] but constructed the APWP for Eurasia using no Asian data in order to avoid any circular reasoning. Data from North America, Africa and India were transferred to Eurasia using ocean kinematics. The resulting curve is included in Figures 8 and 9, with circles of 95% confidence shown on Figure 8c. Inclusion of Eurasian data produces an almost identical APWP. The APWP features a Jurassic-Cretaceous hairpin loop, which stalls at its tip from about 130 to 80 Ma. Around 70 Ma the pole begins to wander along a new Tertiary track, but it is not until 60 Ma that the pole moves significantly away from the stall point.

The paleomagnetic directions isolated in the Sichuan Basin roughly fit the stall point, as is clear in Figure 8, but to compare the directions by age, we have converted the APWP to the direction expected at the sampling site (103°E, 30°N) (Figure 10).

First, we must consider the possibility of a regional prefolding remagnetization because the directions we determined are so stable throughout all the formations sampled. Regional remagnetizations are thought to be the result of fluids circulating through the rocks during orogeny [e.g., Oliver, 1986]. In Sichuan, remagnetization would probably be related to the Tertiary India-Asia collision. Regardless of when this remagnetization could have taken place in the Tertiary, the discrepancy between the paleolatitude predicted by the APWP and that of Sichuan implies at least 15° (~1500 km) of northward shortening. There is no evidence of such large motion between South China and Siberia in such recent times. This fact, taken together with the presence of both polarities (including one section with a correlatable magnetostratigraphy), argue against the hypothesis of a regional prefolding remagnetization.

As mentioned before, the ages of the formations sampled are uncertain. If the Upper Jurassic sites are representative of the paleomagnetic field back to 152 Ma, and the lower Tertiary sites right up to the end of the Eocene (36 Ma), then the paleolatitude plot (Figure 10a) indicates that there was a north-south gap of about 15° (~1500 km) at the beginning and end of this interval. In other words, if the direction we isolated is truly representative of the Sichuan position until the end of the Eocene, then we have the same problem of a missing collision as in the case of a Tertiary remagnetization. We are sceptical of this interpretation, since there is no geological evidence of such a large recent collision between South China and Siberia. In any case, the lower Tertiary result is based on only two sites from one locality, so it cannot be considered as reliable as those with other ages.

The Upper Cretaceous age Nanjing pole [Kent, et al., 1986] is significantly different in paleolatitude from other similar aged poles from China but is in excellent agreement with the Eurasian APWP for 60 Ma (Figure 9). Possibly, this pole corresponds to a magnetization acquired during the northward motion of the region during the early Tertiary and thus postdates all other Chinese poles dated to this period.

The paleolatitude gap in the Late Jurassic may be significant, indicating that collision took place at that time. Paleomagnetism has shown that North and South China are clearly separate from Eurasia in the Upper Permian and

Lower Triassic [e.g., Opdyke et al., 1986], but there is no data for poles between that time and the end of the Jurassic. Reliable poles from the Upper Triassic and Lower Jurassic are necessary to resolve this important problem.

Our preferred interpretation is that the magnetizations determined in the present study were acquired during the Cretaceous (~130-70 Ma), and thus the time span covered is slightly less than indicated by the assigned ages (but never in contradiction with the paleontological and magneto-

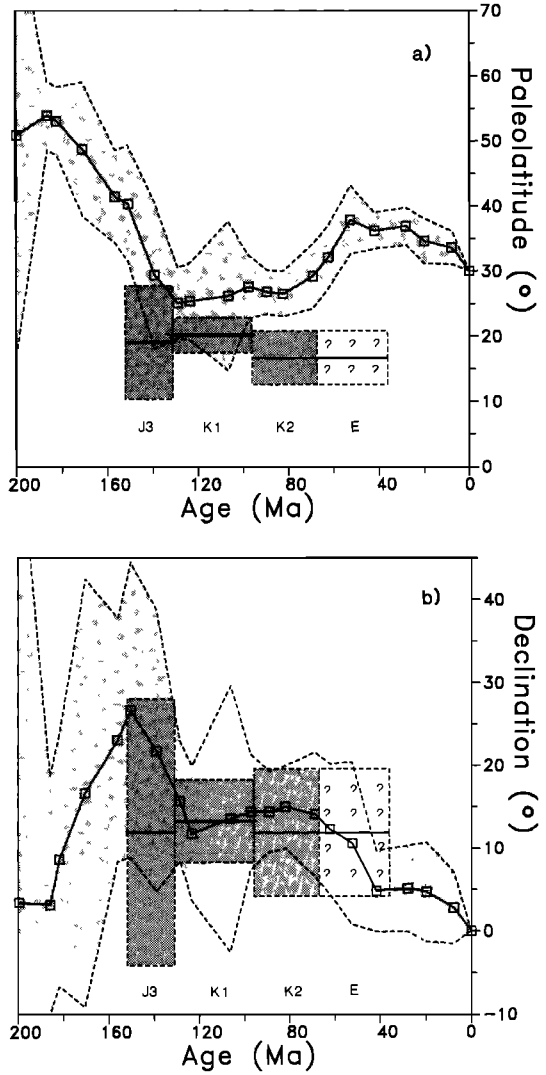


Fig. 10. The (a) paleolatitude and (b) paleodeclination of the Sichuan Basin versus age, compared to that predicted by the Eurasian apparent polar wander path (APWP). Eurasian values are consistent with the poles shown in Figures 8 and 9 and are computed for a reference site at 30°N, 103°E. The Sichuan Basin directions are given as Upper Jurassic, Lower Cretaceous and Upper Cretaceous averages consistent with poles shown in Figure 8a. The area marked with question marks marks the extension of the Upper Cretaceous average into the lower Tertiary, which we consider unlikely (see text).

stratigraphic age constraints). Even so, the paleolatitude of South China is 8.0°±4.0° (at 90 Ma) or 6.6°±6.3° (at 120 Ma) lower than that expected if it were rigidly attached to Eurasia in its present configuration. We would be tempted to interpret this northward shortening as the result of some amount of recent thrusting along the Qinling Shan, the mountain range separating South and North China. On the other hand, poles determined from localities north of the Qinling Shan (North China Block, Tarim and Junggar) also indicate the same anomalously southerly position (Figure 9; poles are on the far side of the APWP with respect to China).

While the uncertainties of individual studies make these poles indistinguishable with the Eurasian stall point, the trend is significant when they are taken together. Either a few hundred kilometers of northward motion must have been accommodated somewhere between Europe and China during the Tertiary (as a result of the Indian collision?) or the reference APWP is still inaccurate.

The paleodeclination of the Sichuan pole fits very well with the Eurasian APWP (Figure 10b) as do the poles from Zhejiang (South China Block [Lin et al., 1986]) and the Junggar Block [Chen et al., 1991]. In contrast, poles from Shandong [Lin et al., 1985] and Korea [Lee et al., 1987] on or east of the Tanlu Fault (see Figure 1), and poles from the Tarim block [Li et al., 1988] seem to have rotated about 10° clockwise. The latter rotation is considered by Chen et al. [1991] to be a consequence of the India-Asia collision.

CONCLUSION

We have presented paleomagnetic results from three Cretaceous localities in the Sichuan Basin separated by up to 150 km. Each locality alone has features to help prove the reliability of the result, but each in its own way has a weakness.

The Feixianguan region (reported in detail in Enkin et al., [1991]) has many sites (26 accepted out of 28 sampled), a magnetostratigraphically determined age, a 99% significant fold test, both polarities, and a well-determined average direction. Unfortunately, the whole section has suffered a local rotation, which we estimate with two different methods using geological constraints alone.

The four sites from around Xinjin have a widely dispersed low-temperature component, but their high-temperature components are well defined and are clustered better on structural correction (although not significantly with so few sites). Both polarities are represented.

The demagnetizations of the samples from the Guanyin locality are particularly clean from noise, but the site averages are quite dispersed, the series is completely magnetized in normal polarity, and the beds are flat lying so that no fold test is possible. An isolated older site some 50 km away contains the antipodal direction.

The reassuring feature of this study is that the locality directions are identical, so all the stability tests can be taken together to make one reliable primary result. It is unlikely that a remagnetization event could affect an extended region while not annulling any stability tests. (Recall, for instance, that the pervasive North American Kiaman remagnetization is consistently in reversed polarity, and thus no reversals are

observed in affected rocks.) Furthermore, a regional prefolding remagnetization event during the Tertiary should have given a significantly steeper direction than is measured.

We are surprised to note that the paleomagnetic direction is so stable with time. On the other hand, the reported ages largely overlap with the duration of the stable tip of the Eurasian apparent polar wander path, as defined with poles from other continents transferred onto Eurasia. Accordingly, we interpret our results to represent only the period of the stable tip (~130 - ~70 Ma), shorter than the assigned ages for the formations we sampled but not in contradiction with paleontological and magnetostratigraphic age constraints.

The same results are evident in the poles of the other major Chinese blocks. In no other place in China, however, have such a large number of samples of this age been studied and such a battery of stability tests been applied. This study in Sichuan lends credence to the other poles in the literature.

The Chinese Cretaceous poles together suggest that the Chinese mosaic was fully assembled by the Late Jurassic and subsequent motion followed the rest of "stable" Eurasia to the present, except for small (<10°) relative rotations and possibly a few hundred kilometers of northward shortening in Siberia.

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